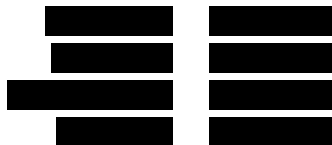


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Project Specifications and Risk Assessment
Automated Parking Payment System

Group 2018.018



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Table of Contents

1 High-level Description	2
1.1 Motivation.....	2
1.2 Project Objective.....	2
1.3 Block Diagram	2
2 Project Specifications.....	4
2.1 Functional Specifications.....	4
2.2 Non-functional Specifications.....	6
3 Risk Assessment	7
References	9

1 High-level Description

1.1 Motivation

Each year, approximately 2.8 million parking tickets are issued across the City of Toronto [1]. A large fraction of these arises from the inconvenience of finding and correctly using manual parking meters as well as from underestimating parking time. Many drivers find the current payment process to be cumbersome and would prefer a more seamless method. An ideal parking payment system does not require drivers to estimate parking duration, to bring payment, to find a parking meter and correctly use it, and to select the correct parking rate.

Automated Parking Payment System (APPS) can reduce the number of parking tickets as well as the inconveniences associated with parking payment by introducing a fairer system that automatically charges parked cars for their precise usage, rather than pre-billing drivers and punishing those who do not adhere to their estimated time.

1.2 Project Objective

The objective is to design an automatic parking payment system that charges the driver of parked cars for their precise usage based on real-time parking information.

1.3 Block Diagram

The proposed solution is a system that consists of readers on parking posts which communicate with tags installed inside cars, tracking the time that a car is parked. The system architecture is illustrated in **Figure 1**, and consists of five main subsystems: tag & reader communication, microcontroller, central controller, payment system, and backend database servers.

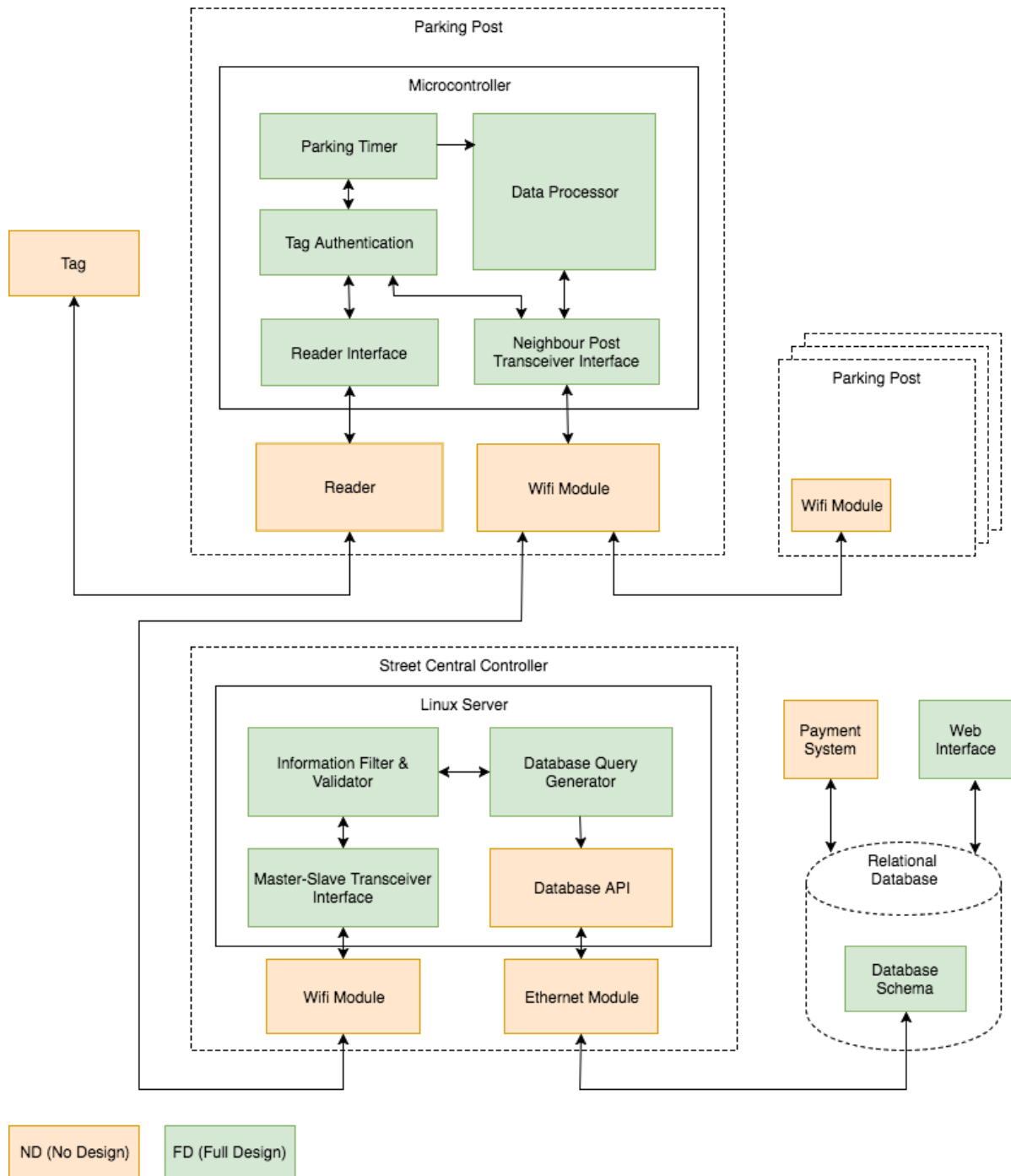


Figure 1. Block diagram of the Automatic Parking Payment System

Table 1 provides a more detailed description of the components shown in **Figure 1**.

Table 1. Description of block diagram components

Component	To be designed	Description
Tag	No	Each tag contains a unique identification number (UID) that correlates to an account in the system. The UID is stored on the tag upon manufacture.
Reader	No	The reader scans for tags and retrieves the UID stored in the tag.
Microcontroller	No	The microcontroller hosts the reader and communicates with other microcontrollers wirelessly. Some also communicate directly with the central controller.
Firmware	Yes	The firmware on the microcontroller will periodically send commands to the reader to scan for tags. It will keep track of parking duration and report to the central controller at every car departure.
Authentication Process	Yes	The central controller will check tag UID against database to retrieve the associated user information
Web interface	Yes	Allows users to register and to view their transaction history
Payment system	No	This system securely charges user's bank account after each car park
Database Server	Yes	Stores information regarding user, bank account, transaction history

2 Project Specifications

Project specifications are classified into functional and non-functional specifications.

2.1 Functional Specifications

Table 2 lists the functional specifications identified in the project along with their descriptions and their level of necessity.

Table 2. Functional specifications

Subsystem	Specification	Description	Necessity
Reader & tag	Reader-tag Communication range	Each reader shall communicate wirelessly with one or more tags within a minimum 10 m radius	Essential
Micro-Controller	Microcontroller communication range	A parking post shall communicate with another parking post within a minimum 15 m radius	Essential
	Network Connectivity	Multiple parking posts shall be able to form an Ad-hoc network that connects to the central controller	Essential
	Peripherals	The microcontroller shall support wireless communication between parking posts	Essential
	Time Tracking	The microcontroller shall record arrival/departure time of each tag within its reader's range	Essential
	Transaction Reporting	The microcontroller shall report to the central controller after each departure of a tag	Essential
Central Controller	Conflict resolution	The central controller shall resolve any conflicts or duplicate information	Essential
	Microcontroller communication	The central controller shall process information coming from all microcontrollers connected to it	Essential
	Transaction Recording	The central controller shall record transactions to remote database servers	Essential
Database	Transaction Storage	The database shall permanently store all transaction records	Essential
Payment System	Payment	Users shall be made available a payment method for the amount charged	Essential
Application	User registration	They system shall support user registration	Essential
	User notification	The system shall notify users of any transactions that occur on his/her	Non-Essential

	bank account	
Transaction history	The system shall provide an interface for users to view their transaction history	Essential
Parking time accuracy	The system shall charge users by the minute	Essential
Monthly parking	The system shall provide monthly pass services	Non-essential
Pay-as-you-go parking	The system shall provide pay-as-you-go services	Essential

2.2 Non-functional Specifications

Some design specifications do not directly relate to how the system operates; instead, they describe the characteristics of the system. These non-functional specifications are listed in **Table 3**.

Table 3. Non-functional specifications

Subsystem	Specification	Description	Necessity
Tag	Tag location	Each tag shall be attached to the car windshield, where it is visible	Essential
	Tag ID	Each tag shall contain a unique encrypted identification number	Essential
	Security	Communication between tags and readers shall be encrypted	Essential
	Tag cost	Each tag shall cost less than \$2 CAD to produce	Non-essential
	Tag size	Each tag shall be less than 5 cm by 5 cm by 1 cm in volume	Non-essential
Micro-Controller	Microcontroller/reader cost	Each set of microcontroller and reader shall cost less than \$50 to produce	Non-essential
	Microcontroller/reader Power supply	A parking post shall receive continuous, uninterrupted power	Essential

	Information preservation	The microcontroller shall retain all information upon power failure	Non-essential
	Power Consumption	The microcontroller shall consume less than 10 W	Non-essential
	Memory capacity	The microcontroller shall have at least 16 kB flash memory	Essential
Central Controller	Power requirement	The central controller shall receive continuous, uninterrupted power	Essential
	Throughput	The central controller shall handle up to 100 microcontrollers	Non-essential
Database Servers	Backend server uptime	The database servers shall have an uptime of 99%	Essential
Overall	Reliable data transmission	The overall system shall ensure reliable data transmission among different entities	Essential

3 Risk Assessment

Table 4 shows three risks that could prevent successful completion of the project.

Table 4. Risk Assessment

Risk: System Integration
Impact: High
Probability: High
Situation The ad-hoc parking post network is composed of multiple microcontrollers communicating with each other through some wireless protocols. Since this project is heavily dependent on off-the-shelf hardware parts and third party software (e.g. operating system on the microcontrollers), the selected combination of hardware/software components may not integrate well together and could lead to an impractical solution.
Remediation To lower the probability of this risk, the team should create more production stages and prioritize functionality over scalability, performance, and other bonus features. Always build a prototype for proof-of-concept before making final decisions for any hardware or software acquisitions. Integration of subsystems should be done incrementally; the integration of one subsystem with another should be tested thoroughly before introducing the next subsystem.

Risk: Insufficient Funds

Impact: Medium

Probability: Medium

Situation

To fully verify scalability and fault tolerance of the distributed system, a large number of devices will be required for benchmarking and stress tests. However, if funding is not sufficient and thus not enough devices are available for verification, there will be no way to demonstrate whether the project has met all the requirements.

Remediation

Plan ahead: collect as much funding as possible through design project awards and sponsorship. Borrow hardware devices instead of purchasing them out of the pocket from the consultant's research group.

Risk: Time Management

Impact: High

Probability: Low

Situation

Aside from ECE498, the group members are taking challenging technical electives and participating in job interviews which are time consuming. Thus, conflicting schedules will arise and coordinating will become difficult. It is likely that some members will not be able to attend all the group meetings, in which case part of the team must make decisions for the entire team. If groupmates absent from the meeting do not agree on such decisions, conflicting ideas may halt progress and the team may further feel demotivated to carry on. Success of the project will be jeopardized due to low team spirits.

Remediation

Keep a clean list of deliverables with associated deadlines so that everyone on the team can prioritize FYPD among all other tasks. Also, be flexible with scheduling group meetings to achieve highest attendance rate possible. Teammates should value each other's opinions and inputs to the project.

References

- [1] "Parking Tickets - Data catalogue - Open Data | City of Toronto", *www1.toronto.ca*, 2017. [Online]. Available:
<http://www1.toronto.ca/wps/portal/contentonly?vgnextoid=ca20256c54ea4310VgnVCM1000003dd60f89RCRD>. [Accessed: 29- May- 2017].